

121. Let $y = y(x)$ be the solution of the differential equation $\sec x dy + \{2(1-x) \tan x + x(2-x)\} dx = 0$ such that $y(0) = 2$. Then $y(2)$ is equal to :

माना अवकल समीकरण $\sec x dy + \{2(1-x) \tan x + x(2-x)\} dx = 0$, $y(0) = 2$ का हल $y = y(x)$ है। तो $y(2)$ बराबर है :

- (A) 1 (B) $2\{\sin(2) + 1\}$ (C) 2 (D) $2\{1 - \sin(2)\}$

Ans. C

Sol. $dy = \frac{2(x-1) \tan x + x(x-2)}{\sec x} dx$

$$\Rightarrow \int dy = \int 2(x-1) \sin x + (x^2 - 2x) \cos x$$

$$\Rightarrow y = 2[(x-1)(-\cos x) - (1)(-\sin x)] + [(x^2 - 2x)(\sin x) - (2x-2)(-\cos x) + (2)(-\sin x)] + c$$

$$y(0) = 2[1] + [-2] + C$$

$$y(0) = 2$$

$$C = 2$$

$$y(2) = 2(1 + \sin 2) - 2\sin 2 = 2$$

122. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x (\sec x - \sin x \tan x)}$, $x \in \left(0, \frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right) = 2$. Then $y\left(\frac{\pi}{3}\right)$, is :

माना अवकल समीकरण $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x (\sec x - \sin x \tan x)}$, $x \in \left(0, \frac{\pi}{2}\right)$ के हल $y = y(x)$ के लिए $y\left(\frac{\pi}{4}\right) = 2$ है। तो

$y\left(\frac{\pi}{3}\right)$ बराबर है :

- (A) $\sqrt{3}(2 + \log_e \sqrt{3})$ (B) $\frac{\sqrt{3}}{2}(2 + \log_e 3)$ (C) $\sqrt{3}(2 + \log_e 3)$ (D) $\sqrt{3}(1 + 2 \log_e 3)$

Ans. A

Sol. $\frac{dy}{dx} = \frac{\sin x + y \cos x}{\sin x \cdot \cos x \left(\frac{1}{\cos x} - \sin x \frac{\sin x}{\cos x} \right)}$

$$= \frac{\sin x + y \cos x}{\sin x (1 - \sin^2 x)}$$

$$\frac{dy}{dx} = \sec^2 x + y \cdot 2(\operatorname{cosec} 2x)$$

$$\frac{dy}{dx} - 2\operatorname{cosec}(2x) \cdot y = \sec^2 x$$

$$\frac{dy}{dx} + p \cdot y = Q$$

$$\text{I.F.} = e^{\int p dx} = e^{\int -2\operatorname{cosec}(2x) dx}$$

$$\text{Let } 2x = t$$

$$2 \frac{dx}{dt} = 1$$

$$dx = \frac{dt}{2}$$

$$= e^{-\int \operatorname{cosec}(t) dt}$$

$$= e^{-\ell n \left| \tan \frac{t}{2} \right|}$$

$$= e^{-\ell n |\tan x|} = \frac{1}{|\tan x|}$$

$$y(\text{IF}) = \int Q(\text{IF}) dx + c$$

$$\Rightarrow y \frac{1}{|\tan x|} = \int \sec^2 x \cdot \frac{1}{|\tan x|} + c$$

$$y \cdot \frac{1}{|\tan x|} = \int \frac{dt}{|t|} + c$$

$$\text{for } \tan x = t$$

$$y \cdot \frac{1}{|\tan x|} = \ell n |t| + c$$

$$y = |\tan x| (\ell n |\tan x| + c)$$

$$\text{Put } x = \frac{\pi}{4}, y = 2$$

$$2 = \ell n 1 + c \Rightarrow c = 2$$

$$y = |\tan x| (\ell n |\tan x| + 2)$$

$$y\left(\frac{\pi}{3}\right) = \sqrt{3}(\ln\sqrt{3} + 2)$$

123. The solution curve of the differential equation $y \frac{dx}{dy} = x(\log_e x - \log_e y + 1)$, $x > 0$, $y > 0$ passing through the point $(e, 1)$ is :

बिन्दु $(e, 1)$ से होकर जाने वाला, अवकल समीकरण $y \frac{dx}{dy} = x(\log_e x - \log_e y + 1)$, $x > 0$, $y > 0$ का हल वक्र है :

(A) $\left| \log_e \frac{y}{x} \right| = y^2$ (B) $\left| \log_e \frac{y}{x} \right| = x$ (C) $2 \left| \log_e \frac{x}{y} \right| = y + 1$ (D) $\left| \log_e \frac{x}{y} \right| = y$

Ans. D

Sol. $y \frac{dx}{dy} = x(\ln x - \ln y + 1)$

$$\frac{dx}{dy} = \frac{x}{y} \left(\ln \frac{x}{y} + 1 \right)$$

Let $x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$

$$\Rightarrow v + y \frac{dv}{dy} = v \ln v + v \Rightarrow y \frac{dv}{dy} = v \ln v \Rightarrow \int \frac{dv}{v \ln v} = \int \frac{dy}{y}$$

$$\Rightarrow \ln(\ln v) = \ln y + \ln c \Rightarrow \ln v = y + c \Rightarrow \ln \left(\frac{x}{y} \right) = y + c$$

passes through $(e, 1) \Rightarrow \ln e = c \Rightarrow c = 1 \Rightarrow y = \ln \left(\frac{x}{y} \right) \Rightarrow y = \left| \ln \left(\frac{x}{y} \right) \right|$ ($\because x > 0, y > 0$)

124. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$, $y(0) = 1$. Then,

$$\left(\frac{1}{\sqrt{2}} + y \left(\frac{1}{\sqrt{2}} \right) \right)^2 \text{ equals :}$$

माना अवकलन समीकरण $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$, $y(0) = 1$ का हल $y = y(x)$ है। तो $\left(\frac{1}{\sqrt{2}} + y \left(\frac{1}{\sqrt{2}} \right) \right)^2$ बराबर

है :

(A) $\frac{4}{4+\sqrt{e}}$

(B) $\frac{3}{3-\sqrt{e}}$

(C) $\frac{1}{2-\sqrt{e}}$

(D) $\frac{2}{1+\sqrt{e}}$

Ans. C

Sol. $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$

Put $x + y = t$

$$\Rightarrow \frac{dy}{dx} = \frac{dt}{dx} - 1$$

$$\frac{dt}{dx} - 1 = 2x(t)^3 - xt$$

$$\Rightarrow \frac{dt}{2t^3 - t} = x dx$$

$$\int \frac{1}{2t^3 - t} dt = \int x dx$$

$$\Rightarrow \int \frac{t}{2t^4 - t^2} dt = \int x dx$$

$$t^2 = z$$

$$2t dt = dz$$

$$\frac{1}{2} \int \frac{dz}{2z^2 - z} = \int x dx$$

$$\ln \left| \frac{z - \frac{1}{2}}{z} \right| = x^2 + c$$

$$\ln \left| \frac{(x+y)^2 - \frac{1}{2}}{(x+y)^2} \right| = x^2 + c$$

$$y(0) = 1 \Rightarrow c = \ln \left(\frac{1}{2} \right)$$

$$\Rightarrow \frac{(x+y)^2 - \frac{1}{2}}{(x+y)^2} = e^{x^2} \times \frac{1}{2}$$

$$\Rightarrow \frac{(x+y)^2 - \frac{1}{2}}{(x+y)^2} = \sqrt{e} \times \frac{1}{2}$$

$$\Rightarrow (x+y)^2 = \frac{1}{2-\sqrt{e}}$$

125. Let α be a non-zero real number. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f(0) = 2$ and $\lim_{x \rightarrow -\infty} f(x) = 1$.

If $f'(x) = \alpha f(x) + 3$, for all $x \in \mathbb{R}$, then $f(-\log_e 2)$ is equal to :

माना α एक शून्येतर वास्तविक संख्या है। माना एक अवकलनीय फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ के लिए $f(0) = 2$ तथा $\lim_{x \rightarrow -\infty} f(x) = 1$ हैं। यदि

$f'(x) = \alpha f(x) + 3$, $x \in \mathbb{R}$ है, तो $f(-\log_e 2)$ बराबर है :

- (A) 9 (B) 5 (C) 7 (D) 3

Ans. Bonus

Sol. $f(0) = 2$, $\lim_{x \rightarrow -\infty} f(x) = 1$

$$f'(x) = \alpha f(x) + 3$$

$$\frac{dy}{dx} = \alpha y + 3$$

$$\int \frac{dy}{\alpha y + 3} = \int dx$$

$$\frac{1}{\alpha} \ln(\alpha y + 3) = x + C$$

$$\alpha y + 3 = ce^{\alpha x}$$

$$\because \lim_{x \rightarrow -\infty} f(x) = 1 \Rightarrow \alpha > 0 \text{ but } \alpha = -3$$

So given data contradicts the information

126. Let $y = y(x)$ be the solution of the differential equation $(1-x^2)dy = [xy + (x^3 + 2)\sqrt{3(1-x^2)}]dx$,

$-1 < x < 1, y(0) = 0$. If $y\left(\frac{1}{2}\right) = \frac{m}{n}$, m and n are co-prime numbers, then $m + n$ is equal to _____.

माना अवकल समीकरण $(1-x^2)dy = [xy + (x^3 + 2)\sqrt{3(1-x^2)}]dx$, $-1 < x < 1, y(0) = 0$, का हल $y = y(x)$ है। यदि

$y\left(\frac{1}{2}\right) = \frac{m}{n}$ है, m तथा n असहभाज्य संख्याएँ हैं, तो $m + n$ बराबर है _____।

Ans. 97

Sol.
$$\frac{dy}{dx} = \frac{xy + (x^3 + 2)\sqrt{3(1-x^2)}}{(1-x^2)}$$

$$\frac{dy}{dx} = \frac{xy}{1-x^2} + \frac{(x^3 + 2)\sqrt{3}}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} - \frac{xy}{1-x^2} = \frac{\sqrt{3}(x^3 + 2)}{\sqrt{1-x^2}}$$

$$\text{I.F.} = e^{-\int \frac{x}{1-x^2} dx}$$

$$\Rightarrow e^{-\frac{1}{2} \int \frac{2x}{1-x^2} dx}$$

$$\Rightarrow e^{\frac{1}{2} \ln|1-x^2|}$$

$$\Rightarrow e^{\ln\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}$$

$$\sqrt{1-x^2} \frac{dy}{dx} - \frac{\sqrt{1-x^2}}{-x^2+1} \cdot xy = \sqrt{3}(x^3 + 2)$$

$$\sqrt{1-x^2} \frac{dy}{dx} - \frac{1}{\sqrt{1-x^2}} \cdot xy = \sqrt{3}(x^3 + 2)$$

$$y\sqrt{1-x^2} = \sqrt{3} \int x^3 + 2 dx$$

$$y\sqrt{1-x^2} = \sqrt{3} \left(\frac{x^4}{4} + 2x + c \right)$$

$$y(0) = 0$$

$$\therefore c = 0$$

$$y\sqrt{1-x^2} = \sqrt{3} \left(\frac{x^4}{4} + 2x \right)$$

$$y\left(\frac{1}{2}\right) = ?$$

$$y \frac{\sqrt{3}}{2} = \sqrt{3} \left(\frac{1}{64} + 1 \right)$$

$$y = \frac{65}{32}$$

$$\frac{m}{n} = \frac{65}{32}$$

$$\therefore m + n = 97$$

127. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable odd function satisfying $f'(x) \geq 0$, $f''(x) = f(x)$, $f(0) = 0$, $f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____.

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ एक तीन बार अवकलनीय विषम फलन है, जो $f'(x) \geq 0$, $f''(x) = f(x)$, $f(0) = 0$, $f'(0) = 3$ को संतुष्ट करता है। तो $9f(\log_e 3)$ बराबर है _____.

Ans. 36

Sol. $f''(x) = f(x)$

$$\Rightarrow f'(x) f''(x) = f'(x) \cdot f(x)$$

$$\Rightarrow \frac{(f'(x))^2}{2} = \frac{(f(x))^2}{2} + C$$

$$\Rightarrow (f'(x))^2 = (f(x))^2 + C'$$

$$f(0) = 0, f'(0) = 3 \Rightarrow C' = 9$$

$$\therefore (f'(x))^2 = (f(x))^2 + 9$$

$$f'(x) = \sqrt{(f(x))^2 + 9} \quad \therefore f'(x) \geq 0$$

$$\int \frac{dy}{\sqrt{y^2 + 9}} = \int dx \Rightarrow \ln \left| y + \sqrt{y^2 + 9} \right| = x + C$$

$$\Rightarrow f(0) = 0 \Rightarrow C = \ln 3$$

$$\Rightarrow y + \sqrt{y^2 + 9} = 3e^x$$

$$\text{at } x = \ln 3; y = 4$$

$$\therefore 9f(\ln 3) = 36$$

128. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 2y \sec^2 x = 2 \sec^2 x + 3 \tan x \cdot \sec^2 x$ such that $y(0) = \frac{5}{4}$.

Then $12\left(y\left(\frac{\pi}{4}\right) - e^{-2}\right)$ is equal to _____.

माना अवकल समीकरण $\frac{dy}{dx} + 2y \sec^2 x = 2\sec^2 x + 3 \tan x \cdot \sec^2 x$, $y(0) = \frac{5}{4}$ का हल $y = y(x)$ है। तो $12\left(y\left(\frac{\pi}{4}\right) - e^{-2}\right)$ बराबर है _____.

Ans. 21

Sol. $\frac{dy}{dx} + 2y \sec^2 x = 2\sec^2 x + 3 \tan x \sec^2 x$

$$\text{I.F.} = e^{\int 2\sec^2 x dx}$$

$$\text{I.F.} = e^{2\tan x}$$

$$y \cdot e^{2\tan x} = \int e^{2\tan x} (2 + 3 \tan x) \sec^2 x dx$$

$$\text{Put } \tan x = u$$

$$\sec^2 x dx = du$$

$$y \cdot e^{2u} = \int e^{2u} (2 + 3u) du$$

$$y \cdot e^{2u} \Rightarrow \frac{2e^{2u}}{2} + 3 \int e^{2u} \cdot u du$$

$$y \cdot e^{2u} = e^{2u} + 3 \left[\frac{ue^{2u}}{2} - \int \frac{e^{2u}}{2} \right]$$

$$ye^{2u} = e^{2u} + 3 \left[\frac{ue^{2u}}{2} - \frac{e^{2u}}{4} \right] + C$$

$$ye^{2\tan x} = e^{2\tan x} + 3 \left[\frac{\tan x e^{2\tan x}}{2} - \frac{e^{2\tan x}}{4} \right] + C$$

$$F(0) = \frac{5}{4}$$

$$\frac{5}{4} = 1 - \frac{3}{4} + C$$

$$\frac{5}{4} - \frac{1}{4} = C$$

$$1 = C$$

$$y = 1 + 3\left(\frac{\tan x}{2} - \frac{1}{4}\right) + 1 \cdot e^{-2 \tan x}$$

$$y\left(\frac{\pi}{4}\right) = 1 + 3\left(\frac{1}{2} - \frac{1}{4}\right) + \frac{1}{e^2}$$

$$y\left(\frac{\pi}{4}\right) = \frac{7}{4} + \frac{1}{e^2}$$

$$12\left(y\left(\frac{x}{4}\right) - \frac{1}{e^2}\right) = 12\left(\frac{7}{4} + \frac{1}{e^2} - \frac{1}{e^2}\right) = 21$$

129. Let $y = f(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^6 + 4x}{\sqrt{1 - x^2}}$, $-1 < x < 1$ such that $f(0) = 0$. If

$$6 \int_{-1/2}^{1/2} f(x) dx = 2\pi - \alpha \text{ then } \alpha^2 \text{ is equal to } \underline{\hspace{2cm}}.$$

माना कि $y = f(x)$, अवकल समीकरण $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^6 + 4x}{\sqrt{1 - x^2}}$, $-1 < x < 1$ का हल इस प्रकार है, कि $f(0) = 0$ है। यदि

$$6 \int_{-1/2}^{1/2} f(x) dx = 2\pi - \alpha \text{ है, तो } \alpha^2 \text{ बराबर है } \underline{\hspace{2cm}}।$$

Ans. 27

Sol. $\frac{dy}{dx} + \frac{x}{\underbrace{x^2 - 1}_P} y = \frac{x^6 + 4x}{\underbrace{\sqrt{1 - x^2}}_Q}, -1 < x < 1$

$$\text{I.F.} = e^{\int \frac{x}{x^2 - 1} dx}$$

$$\text{I.F.} = e^{\frac{1}{2} \int \frac{2x}{x^2 - 1} dx} = e^{\frac{1}{2} \ln |x^2 - 1|} = \sqrt{1 - x^2}$$

$$\Rightarrow y \times \sqrt{1 - x^2} = \int \left(\frac{x^6 + 4x}{\sqrt{1 - x^2}} \times \sqrt{1 - x^2} \right) dx$$

$$y \times \sqrt{1 - x^2} = \left(\frac{x^7}{7} + 2x^2 \right) + C$$

$$x = 0$$

$$y = 0$$

$$0 = 0 + C$$

$$C = 0$$

$$y = \frac{x^7}{7\sqrt{1-x^2}} + \frac{2x^2}{\sqrt{1-x^2}} = f(x)$$

$$\Rightarrow 6 \int_{-\frac{1}{2}}^{\frac{1}{2}} f(x) dx = 6 \times \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\underbrace{\frac{x^7}{7\sqrt{1-x^2}}}_{\text{odd}} + \underbrace{\frac{2x^2}{\sqrt{1-x^2}}}_{\text{even}} \right) dx$$

$$\Rightarrow 6 \times 2 \times \int_0^{\frac{1}{2}} \frac{2x^2}{\sqrt{1-x^2}} dx$$

$$\Rightarrow 6 \times 2 \times \int_0^{\frac{1}{2}} \frac{2x^2}{\sqrt{1-x^2}} dx$$

$$= 24 \int_0^{\frac{1}{2}} \left(\frac{x^2 - 1 + 1}{\sqrt{1-x^2}} \right) dx$$

$$= 24 \left(- \int_0^{\frac{1}{2}} \sqrt{1-x^2} + \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} dx \right)$$

$$= 24 \left(- \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x \right]_0^{\frac{1}{2}} + \left[\sin^{-1} x \right]_0^{\frac{1}{2}} \right)$$

$$= 24 \left(- \left(\frac{\sqrt{3}}{8} + \frac{\pi}{12} - 0 \right) + \frac{\pi}{6} - 0 \right)$$

$$= 2\pi - 3\sqrt{3}$$

$$\alpha = 3\sqrt{3}$$

$$\alpha^2 = 27$$

130. Let f be a differentiable function such that $2(x+2)^2 f(x) - 3(x+2)^2 = 10 \int_0^x (t+2)f(t)dt$, $x \geq 0$. Then $f(2)$ is equal to _____.

माना कि f , एक अवकलनीय फलन इस प्रकार है कि $2(x+2)^2 f(x) - 3(x+2)^2 = 10 \int_0^x (t+2)f(t)dt$, $x \geq 0$ है। तब $f(2)$ बराबर

है _____.

Ans. 19

Sol. Differentiate both sides

$$4(x+2)f(x) + 2(x+2)^2 f'(x) - 6(x+2) = 10(x+2)f(x)$$

$$2(x+2)^2 f'(x) - 6(x+2)f(x) = 6(x+2)$$

$$(x+2) \frac{dy}{dx} - 3y = 3$$

$$\int \frac{dy}{dx} = 3 \int \frac{dx}{x+2}$$

$$\ln(y+1) = 3 \ln(x+2) + C$$

$$(y+1) = C(x+2)^3$$

$$f(0) = \frac{3}{2}$$

$$f(2) = 19$$

131. Let $y = y(x)$ be the solution of the differential equation $2 \cos x \frac{dy}{dx} = \sin 2x - 4y \sin x$, $x \in \left(0, \frac{\pi}{2}\right)$. If

$y\left(\frac{\pi}{3}\right) = 0$, then $y'\left(\frac{\pi}{4}\right) + y\left(\frac{\pi}{4}\right)$ is equal to _____.

माना $y = y(x)$ अवकल समीकरण $2 \cos x \frac{dy}{dx} = \sin 2x - 4y \sin x$, $x \in \left(0, \frac{\pi}{2}\right)$ का एक हल है। यदि $y\left(\frac{\pi}{3}\right) = 0$, तो

$y'\left(\frac{\pi}{4}\right) + y\left(\frac{\pi}{4}\right)$ बराबर है _____.

Ans. 1

Sol. $\frac{dy}{dx} + 2y \tan x = \sin x$

$$\text{I.F. : } e^{\int 2 \tan x dx} = \sec^2 x$$

Solution of D.E.

$$y \sec^2 x = \int \sec^2 x \sin x dx + c$$

$$y \sec^2 x = \sec x + c$$

$$y = \cos x + c \cos^2 x$$

$$y\left(\frac{\pi}{3}\right) = \frac{1}{2} + \frac{c}{4} = 0 \Rightarrow c = -2$$

$$y = \cos x - 2 \cos^2 x$$

$$y(x) = -\sin x + 4 \cos x \sin x$$

$$y'\left(\frac{\pi}{4}\right) + y\left(\frac{\pi}{4}\right) = \frac{-1}{\sqrt{2}} + 4\left(\frac{1}{2}\right) + \frac{1}{\sqrt{2}} - 2\left(\frac{1}{2}\right) = 1$$

132. If $y = y(x)$ is the solution of the differential equation, $\sqrt{4-x^2} \frac{dy}{dx} = \left(\left(\sin^{-1} \left(\frac{x}{2} \right) \right)^2 - y \right) \sin^{-1} \left(\frac{x}{2} \right)$, $-2 \leq x \leq 2$,

$$y(2) = \frac{\pi^2 - 8}{4}, \text{ then } y^2(0) \text{ is equal to } \underline{\hspace{2cm}}.$$

यदि $y = y(x)$, अवकल समीकरण, $\sqrt{4-x^2} \frac{dy}{dx} = \left(\left(\sin^{-1} \left(\frac{x}{2} \right) \right)^2 - y \right) \sin^{-1} \left(\frac{x}{2} \right)$, $-2 \leq x \leq 2$, $y(2) = \frac{\pi^2 - 8}{4}$, का हल

है तो $y^2(0)$ बराबर है $\underline{\hspace{2cm}}$ ।

Ans. 4

Sol. $\sqrt{4-x^2} \left(\frac{dy}{dx} \right) = \left(\sin^{-1} \left(\frac{x}{2} \right) \right)^3 - y \sin^{-1} \left(\frac{x}{2} \right)$

$$\left(\frac{dy}{dx} \right) + \frac{y \sin^{-1} \left(\frac{x}{2} \right)}{\sqrt{4-x^2}} = \frac{\left(\sin^{-1} \left(\frac{x}{2} \right) \right)^3}{\sqrt{4-x^2}}$$

$$\Rightarrow P(x) = \frac{\sin^{-1} \left(\frac{x}{2} \right)}{\sqrt{4-x^2}}$$

$$\Rightarrow Q(x) = \frac{\left(\sin^{-1} \left(\frac{x}{2} \right) \right)^3}{\sqrt{4-x^2}}$$

$$\text{I.F.} = e^{\int P(x) dx}$$

$$\int P(x) dx = \int \frac{\sin^{-1}\left(\frac{x}{2}\right)}{\sqrt{4-x^2}} dx$$

$$\sin^{-1}\left(\frac{x}{2}\right) = t$$

$$\frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}} \left(\frac{1}{2}\right) dx = dt$$

$$\left(\frac{1}{\sqrt{4-x^2}}\right) dx = dt$$

$$\int t dt = \frac{t^2}{2}$$

$$e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = \text{I.F.}$$

$$y \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.}) \cdot dx$$

$$y \cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = \int \frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^3}{\sqrt{4-x^2}} \cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} \cdot dx$$

$$\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2} = t$$

$$\frac{\sin^{-1}\left(\frac{x}{2}\right)}{\sqrt{4-x^2}} dx = dt$$

$$y \cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = \int 2t \cdot e^t \cdot dt$$

$$y \cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = 2(t \cdot e^t - e^t) + C$$

$$y \cdot e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} = 2 \left(\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2} - 1 \right) e^{\frac{\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2}{2}} + C$$

$$y(2) = \left(\frac{\pi^2 - 8}{4} \right) \Rightarrow C = 0$$

$$y \Rightarrow \left(\sin^{-1}\left(\frac{x}{2}\right) \right)^2 - 2$$

$$y^2(0) = (-2)^2 = 4$$

133. Let the solution $y = y(x)$ of the differential equation $\frac{dy}{dx} - y = 1 + 4 \sin x$ satisfy $y(\pi) = 1$. Then $y\left(\frac{\pi}{2}\right) + 10$ is equal to _____.

माना अवकल समीकरण $\frac{dy}{dx} - y = 1 + 4 \sin x$ के हल $y = y(x)$ के लिए $y(\pi) = 1$ है। तो $y\left(\frac{\pi}{2}\right) + 10$ बराबर है _____

Ans. 7

Sol. $\frac{dy}{dx} - y = 1 + 4 \sin x$ linear differential equation

$$\text{I.F.} = e^{\int -dx} = e^{-x}$$

So solution of linear differential equation is

$$ye^{-x} = \int e^{-x} (1 + 4 \sin x) dx$$

$$ye^{-x} = -e^{-x} + 4 \int e^{-x} \sin x dx$$

$$ye^{-x} = -e^{-x} + 4 \frac{e^{-x}}{2} [-\sin x - \cos x] + c$$

$$y = -1 - 2(\sin x + \cos x) + ce^x$$

$$\text{Now } y(\pi) = 1$$

$$\Rightarrow 1 = -1 - 2(0 - 1) + ce^\pi$$

$$\text{Hence } y = -1 - 2(\sin x + \cos x)$$

$$\Rightarrow y\left(\frac{\pi}{2}\right) + 10 = -1 - 2(1 + 0) + 10$$

$$= 7$$

134. Let $y = y(x)$ be the solution of the differential equation $(x + y + 2)^2 dx = dy$; $y(0) = -2$. Let the maximum and minimum values of the function $y = y(x)$ in $\left[0, \frac{\pi}{3}\right]$ be α and β , respectively. If $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$, $\gamma, \delta \in \mathbb{Z}$, then $\gamma + \delta$ equals _____.

माना $y = y(x)$ अवकल समीकरण $(x + y + 2)^2 dx = dy$; $y(0) = -2$ का एक हल है। माना $\left[0, \frac{\pi}{3}\right]$ में फलन $y = y(x)$ के उच्चतम तथा निम्नतम मान क्रमशः α तथा β हैं। यदि $(3\alpha + \pi)^2 + \beta^2 = \gamma + \delta\sqrt{3}$, $\gamma, \delta \in \mathbb{Z}$ तो $\gamma + \delta$ बराबर है _____

Ans. 31

Sol. $\frac{dy}{dx} = (x + y + 2)^2$

Put $x + y + 2 = t$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - 1 = t^2$$

$$\tan^{-1} t = x + c$$

$$\tan^{-1} (x + y + 2) = x + c$$

$$\Rightarrow (x = 0, y = -2)$$

$$C = 0$$

$$y = \tan x - x - 2$$

$$\frac{dy}{dx} = \tan^2 x \geq 0$$

$$\forall x \in \left[0, \frac{\pi}{3}\right]$$

$$\Rightarrow y_{\min} = y(0) = -2 = \beta$$

$$\text{And } y_{\max} = y\left(\frac{\pi}{3}\right) = \sqrt{3} - 2 - \frac{\pi}{3} = \alpha$$

$$\text{Now, } (3\alpha + \pi)^2 + \beta^2$$

$$(3\sqrt{3} - 6 - \pi + \pi)^2 + 4$$

$$27 + 36 - 36\sqrt{3} + 4$$

$$67 - 36\sqrt{3}$$

$$\text{So, } \gamma = 67$$

$$\delta = -36$$

$$\Rightarrow \gamma + \delta = 31$$

135. Let f be a differentiable function in the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $2f(2) + 3f(3)$ is equal to _____.

माना अंतराल $(0, \infty)$ में एक अवकलनीय फलन f है, जिसके लिए $f(1) = 1$ है तथा प्रत्येक $x > 0$ के लिए $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$

है। तो $2f(2) + 3f(3)$ बराबर है _____

Ans. 24

Sol. $f(1) = 1$

$$\lim_{x \rightarrow \infty} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$$

$$\lim_{x \rightarrow \infty} \frac{2 + f(x) - x^2 f'(t)}{1} = 1$$

$$= 2xy - x^2 \frac{dy}{dx} = 1$$

$$= \frac{dy}{dx} - \frac{2y}{x} = \frac{-1}{x^2}$$

$$\text{I.f. } e^{-\int \frac{2}{x} dx} = e^{-2 \ln x} = \frac{1}{x^2}$$

$$y \cdot \frac{1}{x^2} = \int \frac{-1}{x^4} dx$$

$$\frac{y}{x^2} = \frac{-x^{-4+1}}{-4+1} + C$$

$$\frac{y}{x^2} = \frac{1}{3x^3} + C$$

$$\because f(1) = 1$$

$$\frac{1}{1} = \frac{1}{3} + C \Rightarrow C = 1 - \frac{1}{3} = \frac{2}{3}$$

$$y = \frac{1}{3x} + \frac{2}{3}x^2$$

$$\text{Now } 2f(2) + 3f(3) = 2 \times \left(\frac{1}{6} + \frac{8}{3} \right) + 3 \times \left(\frac{1}{9} + 6 \right)$$

$$= 2 \times \left(\frac{1+16}{6} \right) + 3 \times \left(\frac{54+1}{9/3} \right) = \frac{17}{3} + \frac{55}{3} = \frac{72}{3} = 24 \text{ Ans.}$$

136. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{2x}{(1+x^2)^2} y = xe^{\frac{1}{(1+x^2)}}$; $y(0) = 0$. Then the area

enclosed by the curve $f(x) = y(x)e^{\frac{1}{(1+x^2)}}$ and the line $y - x = 4$ is _____.

माना अवकल समीकरण $\frac{dy}{dx} + \frac{2x}{(1+x^2)^2} y = xe^{\frac{1}{(1+x^2)}}$; $y(0) = 0$ का हल $y = y(x)$ है। तो वक्र $f(x) = y(x)e^{\frac{1}{(1+x^2)}}$ तथा रेखा

$y - x = 4$ से घिरे क्षेत्र का क्षेत्रफल है _____

Ans. 18

Sol. $\frac{dy}{dx} + \frac{2x}{(1+x^2)^2} y = xe^{\frac{1}{(1+x^2)}}$; $y(0) = 0$

I.F. of linear differential equation,

$$\text{I.F.} = e^{\int \frac{2x}{(1+x^2)^2} dx} = e^{\left(\frac{-1}{1+x^2} \right)}$$

$$\Rightarrow y \left(e^{\left(\frac{-1}{1+x^2} \right)} \right) = \int x \cdot e^{\frac{1}{1+x^2}} \cdot e^{\left(\frac{-1}{1+x^2} \right)} dx$$

$$= \frac{x^2}{2} + c$$

$$\Rightarrow y(0) = 0 \Rightarrow 0(e^{-1}) = c \Rightarrow c = 0$$

$$\Rightarrow y = \frac{e^{\frac{1}{1+x^2}} \cdot x^2}{2}$$

Area between curve $y e^{\left(\frac{-1}{1+x^2}\right)} = \frac{x^2}{2}$ and $y - x = 4$

$$\Rightarrow 2(x+4) = x^2 \Rightarrow x^2 - 2x - 8 = 0$$

$$\Rightarrow (x-4)(x+2) = 0$$

$$\int_{-2}^4 \left[(x+4) - \frac{x^2}{2} \right] dx = \left[\frac{x^2}{2} + 4x - \frac{x^3}{6} \right]_{-2}^4$$

$$= \left(8 + 16 - \frac{64}{6} \right) - \left(2 - 8 + \frac{8}{6} \right)$$

$$= 30 - 12 = 18$$

137. Let a conic C pass through the point (4, -2) and P(x, y), $x \geq 3$, be any point on C. Let the slope of the line touching the conic C only at a single point P be half the slope of the line joining the points P and (3, -5). If the focal distance of the point (7, 1) on C is d, then 12d equals _____.

माना कोई शंकु C बिन्दु (4, -2) से होकर जाता है तथा C पर कोई बिन्दु P(x, y), $x \geq 3$ है। माना शंकु C को मात्र एक बिन्दु P पर स्पर्श करने वाली रेखा की प्रवणता बिन्दु P तथा (3, -5) को मिलाने वाली रेखा की प्रवणता की आधी है। यदि C पर स्थित बिन्दु (7, 1) की नाभीय दूरी d है, तो 12d बराबर है _____

Ans. 75

Sol. P(x, y) and $x \geq 3$

Slope of line at P(x, y) will be $\frac{dy}{dx} = \frac{1}{2} \left(\frac{y+5}{x-3} \right)$

$$\Rightarrow 2 \frac{dy}{(y+5)} = \frac{1}{(x-3)} dx$$

$$\Rightarrow 2 \ln(y+5) = \ln(x-3) + C$$

Passes through (4, -2)

$$\Rightarrow 2 \ln(3) = \ln(1) + C$$

$$\Rightarrow C = 2 \ln(3)$$

$$\Rightarrow 2 \ln(y+5) = \ln(x-3) + 2 \ln(3)$$

$$\Rightarrow 2 \left(\ln \left(\frac{y+5}{3} \right) \right) = \ln(x-3)$$

$$\Rightarrow \left(\frac{y+5}{3} \right)^2 = (x-3)$$

$$\Rightarrow (y+5)^2 = 9(x-3)$$

↓

Parabola

$$4a = 9$$

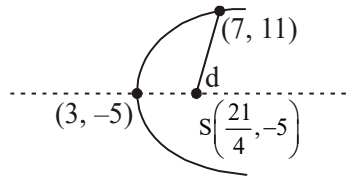
$$a = \frac{9}{4}$$

$$d = \sqrt{\left(\frac{7}{4} \right)^2 + 6^2}$$

$$d = \frac{\sqrt{625}}{4}$$

$$d = \frac{25}{4}$$

$$12d = 75$$



138. If the solution $y(x)$ of the given differential equation $(e^y + 1) \cos x dx + e^y \sin x dy = 0$ passes through the point $\left(\frac{\pi}{2}, 0 \right)$, then the value of $e^{y\left(\frac{\pi}{6}\right)}$ is equal to _____.

यदि अवकल समीकरण $(e^y + 1) \cos x dx + e^y \sin x dy = 0$ का हल $y(x)$ बिन्दु $\left(\frac{\pi}{2}, 0 \right)$ से गुजरता है, तो $e^{y\left(\frac{\pi}{6}\right)}$ का मान बराबर है _____

Ans. 3

Sol. $(e^y + 1) \cos x dx + e^y \sin x dy = 0$

$$\cot x dx + \frac{e^y}{e^y + 1} dy = 0$$

$$\int \cot x dx + \int \frac{e^y}{e^y + 1} dy = 0$$

$$\ln |\sin x| + \ln |e^y + 1| = \ln c$$

$$|\sin x (e^y + 1)| = c$$

$$\text{Put } x = \frac{\pi}{2} \text{ } y = 0$$

$$c = 2$$

$$\sin x (e^y + 1) = 2$$

$$e^y + 1 = 2 \operatorname{cosec} x$$

$$e^y = 2 \operatorname{cosec} x - 1$$

$$e^{y\left(\frac{\pi}{6}\right)} = 2 \operatorname{cosec} \frac{\pi}{6} - 1$$

$$= 2 \times 2 - 1 = 3$$

139. Let $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in \mathbb{N}$ be the solution of the differential equation $xdy - ydx + xy(xdy + ydx) = 0$, $y(1) = 2$. Then $\alpha + \beta$ is equal to _____.

माना अवकल समीकरण $xdy - ydx + xy(xdy + ydx) = 0$, $y(1) = 2$ का हल $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in \mathbb{N}$ है। तो $\alpha + \beta$ बराबर है _____.

Ans. 4

Sol. $\alpha|x| = |y|e^{xy-\beta}$, $\alpha, \beta \in \mathbb{N}$
 $xdy - ydx + xy(xdy + ydx) = 0$

$$\frac{dy}{y} - \frac{dx}{x} + (xdy + ydx) = 0$$

$$\ln|y| - \ln|x| + xy = c$$

$$y(1) = 2$$

$$\ln|2| - 0 + 2 = c$$

$$c = 2 + \ln 2$$

$$\ln|y| - \ln|x| + xy = 2 + \ln 2$$

$$\ln|x| = \ln\left|\frac{y}{2}\right| - 2 + xy$$

$$|x| = \left|\frac{y}{2}\right|e^{xy-2}$$

$$2|x| = |y|e^{xy-2}$$

$$\alpha = 2$$

$$\beta = 2$$

$$\alpha + \beta = 4$$

140. For a differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$, suppose $f'(x) = 3f(x) + \alpha$, where $\alpha \in \mathbb{R}$, $f(0) = 1$ and $\lim_{x \rightarrow -\infty} f(x) = 7$.
Then $9f(-\log_e 3)$ is equal to _____.

एक अवकलनीय फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ के लिए, माना $f'(x) = 3f(x) + \alpha$, जहाँ $\alpha \in \mathbb{R}$, $f(0) = 1$ तथा $\lim_{x \rightarrow -\infty} f(x) = 7$ हैं। तो $9f(-\log_e 3)$ बराबर है _____.

Ans. 61

Sol. $f'(x) = 3f(x) + \alpha$

$$\Rightarrow \frac{dy}{3y + \alpha} = dx$$

$$\Rightarrow \frac{1}{3} \ln(3y + \alpha) = x + C$$

$$y(0) = 1 \Rightarrow C = \frac{1}{3} \ln(3 + \alpha)$$

$$\frac{1}{3} \ln\left(\frac{3y + \alpha}{3 + \alpha}\right) = x$$

$$\Rightarrow y = \frac{1}{3} \left((3 + \alpha)e^{3x} - \alpha \right) = f(x)$$

$$\lim_{x \rightarrow -\infty} f(x) = 7 \Rightarrow \alpha = -21$$

$$\Rightarrow f(x) = 7 - 6e^{3x}$$

$$9f(-\ln 3) = 61$$